

Project Title

ESP32 Smart RFID Attendance Management System with Google Sheets Integration

Objective

Build an advanced attendance management system using ESP32 and RC522 RFID reader. The system should provide attendance logging, a web dashboard, analytics, cloud integration, and administrative features while keeping hardware requirements minimal.

Hardware Components

Mandatory Components

1. ESP32 Development Board
2. RC522 RFID Reader Module
3. RFID Tags/Cards
4. Breadboard and Jumper Wires

Optional Components

1. OLED Display (SSD1306)
 2. DS3231 RTC Module
 3. Buzzer
 4. MicroSD Card Module
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Functionalities to Include in Article

1. User Authentication

- Read RFID card UID
 - Verify registered users
 - Detect unknown cards
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2. Attendance Marking

- Mark attendance automatically
- Prevent duplicate attendance

Example:

John Doe
Already Marked Present

3. Entry and Exit Logging

First scan:

IN : 09:05 AM

Second scan:

OUT : 05:20 PM

4. Working Hours Calculation

Working Hours : 8h 15m

5. Store Records Locally

Using LittleFS/SPIFFS:

- users.json
- attendance.json

No SD card required.

6. Web Dashboard

Display:

- Present users
- Absent users
- Last scanned user
- Entry time
- Exit time

Attendance Table

Show:

Name	Date	IN	OUT	Status
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9. Auto-Refresh Dashboard

Update automatically every few seconds.

No page refresh required.

10. Download CSV Report

Generate:

attendance.csv

Useful for schools and offices.

11. Google Sheets Integration

Every scan automatically uploads:

Name	Date	IN	OUT	Status
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This alone makes the project stand out.

12. Attendance Statistics

Display:

Present : 18

Absent : 7

Attendance Rate : 72%

https://youtu.be/ac5CR_bBg74

<https://youtu.be/ta2wt8-7Xdk>

<https://randomnerdtutorials.com/esp32-rfid-user-management-web-server/>